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A novel digital method for weave pattern recognition based on photometric differential analysis

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ABSTRACT

Based on photometric differential analysis, a new comprehensively-designed and appropriately-performed method was proposed for the pattern analysis of woven fabrics. In this paper, a specially-developed imaging system equipped with a four-side illumination module was established. Firstly, histogram equalization and adaptive wiener filtering for making the yarn profile more apparent were carried out followed by the gradient pyramid fusion of images captured under four illumination directions. Secondly, drawing support from adaptive mesh model, images captured under four illumination directions were divided into sub-images of each interlacing point on the basis of gray scale projection algorithm. The gradation of the convex portion in sub-images was adjusted. Adjusted region was marked as a highlight region to determine attribute category of the interlacing point. Finally, the weave pattern repetition image was generated. Experimental results indicated that almost 97.22% recognition accuracy of weave pattern could be achieved. This method eliminated mutual interference from different yarn directions.

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1. Introduction

As per a specific floating rule, woven fabrics are composed of the warp yarns and weft yarns, forming a certain texture structure. Therefore, because of the basic weave pattern repetition, it makes the weave pattern possess highly-periodic properties. Playing a significant role, fabric weave pattern is absolutely necessary for the textile manufacturing process, in especial, with the advent of high-precision weaving. For instance, it's quite necessary to identify the weave pattern repetition in order to determine the weaving width set-up for the machine and the number of shafts in the preparation and manufacturing process. On the fabric surface, the texture appearance is prone to be affected by the reflected light because of the uneven surface, which is caused by the yarn structure. By means of analyzing the variances of fabric's geometrical surface reflectance, it is available to identify the weave pattern and the texture type of the fabric. In addition, weave pattern recognition methods can be divided into three categories: the template-based method [1,2], the shape-based method [3–6], the statistical texture-based method [7,8]. In recent years, a large number of recognition methods have been proposed by the researchers, to identify the interlacing points.

With the advanced development of computer vision, a variety of great achievements in performing weave pattern repetition detection have been obtained, with the assistance of image processing. Up to now, some scholars have proposed certain methods for automatic recognition of various features of fabrics [9]. In recent studies, the gray scale projection method has been adopted to determine the yarn locations [10,11], in which the image intensity accumulations along the warp and weft projections are classified into peak point or trough point. In the reflected image analysis of woven fabrics, researchers have used the morphological filtering method to determine the locations of the yarn interlacing points. In terms of image segmentation, the binary image should be taken in order to obtain an ideal split image [12]. Due to a slight change happening to environmental factors, the determination of the threshold must be adjusted with the variation of image information, in an automatic manner [13]. Tsai et al. used the gray-level co-occurrence matrix (GLCM) of digital image processing technology to extract the six texture features of fabric defects, which were imported into the back propagation neural network for training and recognizing five fabric defects [14]. With the fuzzy control or a neural network, these intelligent control methods, in usual, are capable of improving the recognition capability [15]. Hu has put forward an automatic fabric classification method based on the

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Bayesian statistics. Kuo et al. used the two-stage Back-Propagation Neural Network to classify the fabric weave patterns. Another identification method [16] was utilized to analyze the warp and weft floats, so as to determine the fabric weave patterns. Very fortunately, these methods mentioned above have been successfully applied to the classification for a wide range of woven fabrics. However, all of the above studies are only on the two-dimensional plane, which will be affected by many factors such as diffuse reflection. By this account, it's a matter of great importance to develop an automatic woven fabric classification method based on photometric differential analysis with the real-time and fault-tolerant functions, to reach the target of replacing traditional visual inspection method.

In fact, the warp and weft yarn information will interfere with each other, at the same time, affect the final identification results [17]. In this case, not only the images captured with illumination directions from the upper side and lower side in the warp direction but also the images captured with illumination directions from the left side and right side in the weft direction can be obtained in a clear form, by controlling the illumination directions [18]. It is a reasonable matter to promote the accuracy of fabric interlacing point recognition by the warp and weft image information obtained based on the photometric differential analysis algorithm [19]. As of today, no related research, able to recognize all fabric textures and create a research bottleneck regarding to the recognition of fabric textures, has been carried out [20]. Dependent on the accurate yarn segmentation, it is an extremely difficult affair for the template-based method and the shape-based method to achieve this objective due to the irregular shape of the yarn floats and the hairiness on the fabric surface [21]. For this reason, the statistical texture-based method was investigated in this paper. Focusing on the appearance of woven fabric, this study is intended for determining the gray-scale features suitable for woven fabric appearance classification so as to establish a novel woven fabric identification system.

Used for weave pattern repetition recognition and based on photometric differential analysis, a novel dedicatedly-developed method was proposed for analyzing the woven fabrics, aiming at obtaining more clear and complete information about the fabric structure. Under the circumstance of the warp yarn information and weft yarn information not interfering with each other, the interlacing point recognition will result in their respective directions with higher precision. With the view of identifying the attributes of interlacing points in an accurate way, the algorithms concerned with image matching and fusion were optimized. Among them, in the photometric stereoscopic imaging system, the relative positions between the camera and target kept unchanged. Only gray scale changes occurred in the images under different illumination conditions, however, without any geometric deviation. Based on the photometric difference, the highlight regions of the protrusions on fabric surface were reconstructed for the weave pattern recognition. Both the highlight region and non-highlight region, used to determine the attribute category of

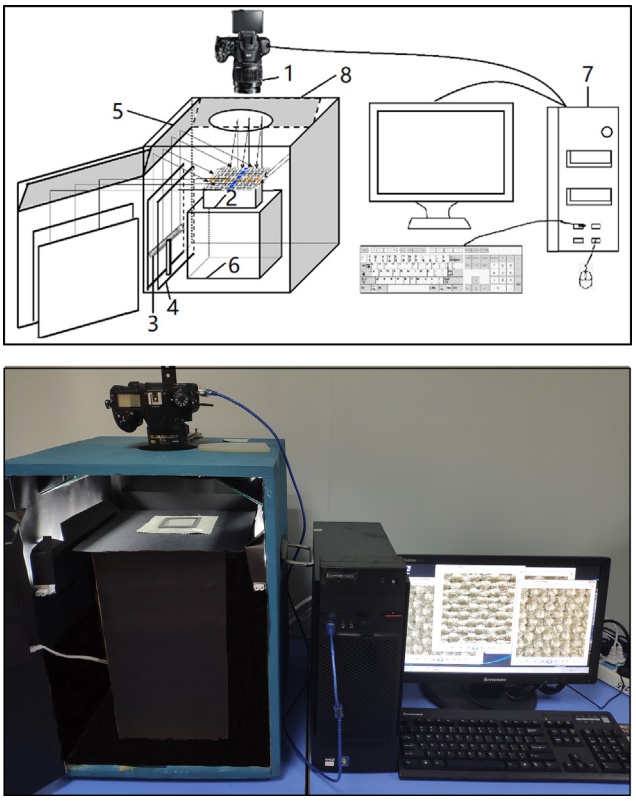


Fig. 2. Image acquisition system based on photometri differential analysis for multiple incident light sources: (1) Camera; (2) anti-"Newton ring" fabric glass fixture; (3) LED parallel light source; (4) LED parallel light source exclusive lampshade; (5) mirror; (6) platform bracket; (7) data analysis server; (8) cabinet.

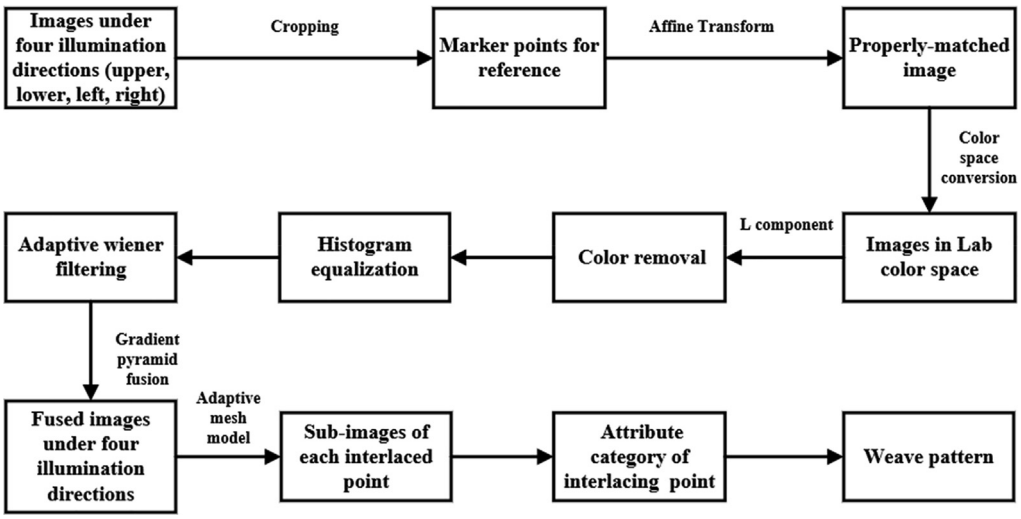


Fig. 1. Flow chart of interlacing point recognition algorithm based on photometric differential analysis.

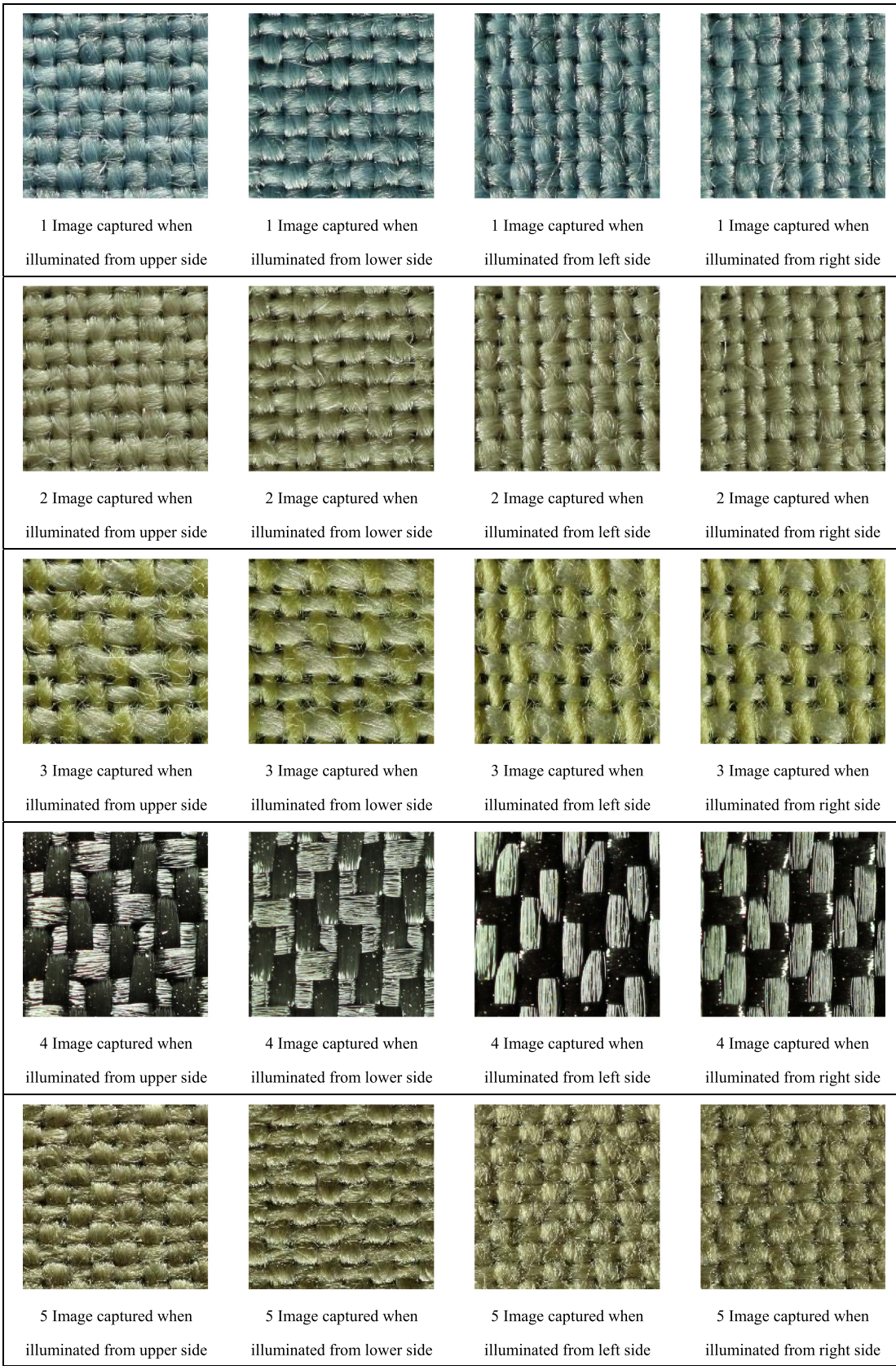


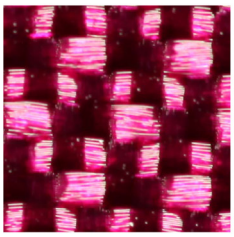
Fig. 3. Matched images of 23 samples captured when illuminated from four directions.

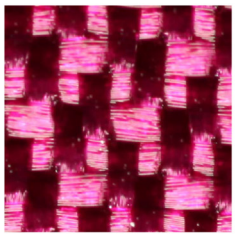

6 Image captured when illuminated from upper side

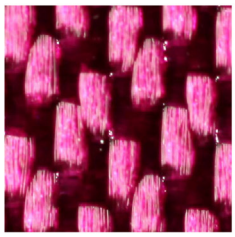

6 Image captured when illuminated from lower side

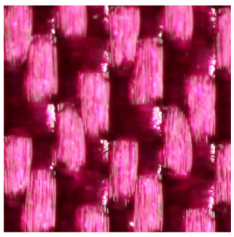

6 Image captured when illuminated from left side


6 Image captured when illuminated from right side


7 Image captured when illuminated from upper side


7 Image captured when illuminated from lower side


7 Image captured when illuminated from left side


7 Image captured when illuminated from right side


8 Image captured when illuminated from upper side


8 Image captured when illuminated from lower side


8 Image captured when illuminated from left side

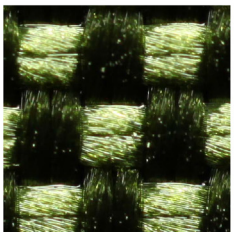

8 Image captured when illuminated from right side

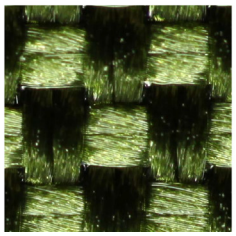

9 Image captured when illuminated from upper side

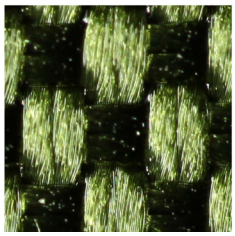

9 Image captured when illuminated from lower side


9 Image captured when illuminated from left side


9 Image captured when illuminated from right side


10 Image captured when illuminated from upper side


10 Image captured when illuminated from lower side


10 Image captured when illuminated from left side

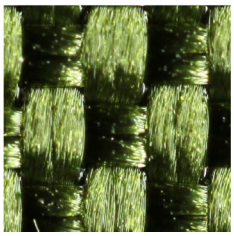

10 Image captured when illuminated from right side

Fig. 3 (*continued*)



11 Image captured when illuminated from upper side



11Image captured when illuminated from lower side



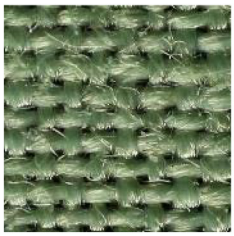
11 Image captured when illuminated from left side



11 Image captured when illuminated from right side



12 Image captured when illuminated from upper side



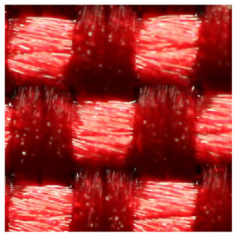
12 Image captured when illuminated from lower side



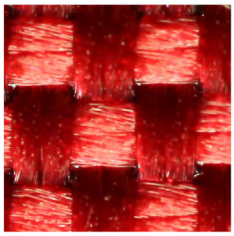
12 Image captured when illuminated from left side



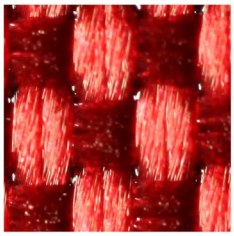
12 Image captured when illuminated from right side



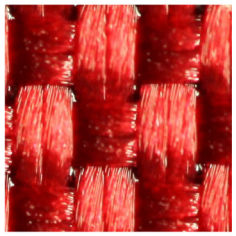
13 Image captured when illuminated from upper side



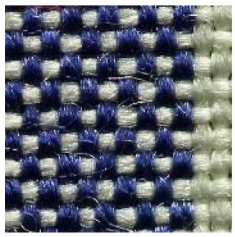
13 Image captured when illuminated from lower side



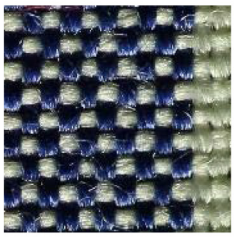
13 Image captured when illuminated from left side



13 Image captured when illuminated from right side



14 Image captured when illuminated from upper side



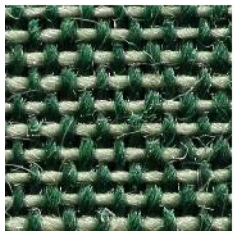
14 Image captured when illuminated from lower side



14 Image captured when illuminated from left side



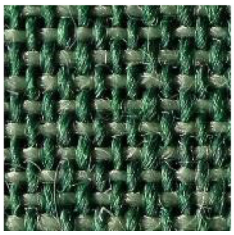
14 Image captured when illuminated from right side



15 Image captured when illuminated from upper side



15 Image captured when illuminated from lower side



15 Image captured when illuminated from left side



15 Image captured when illuminated from right side

Fig. 3 (*continued*)

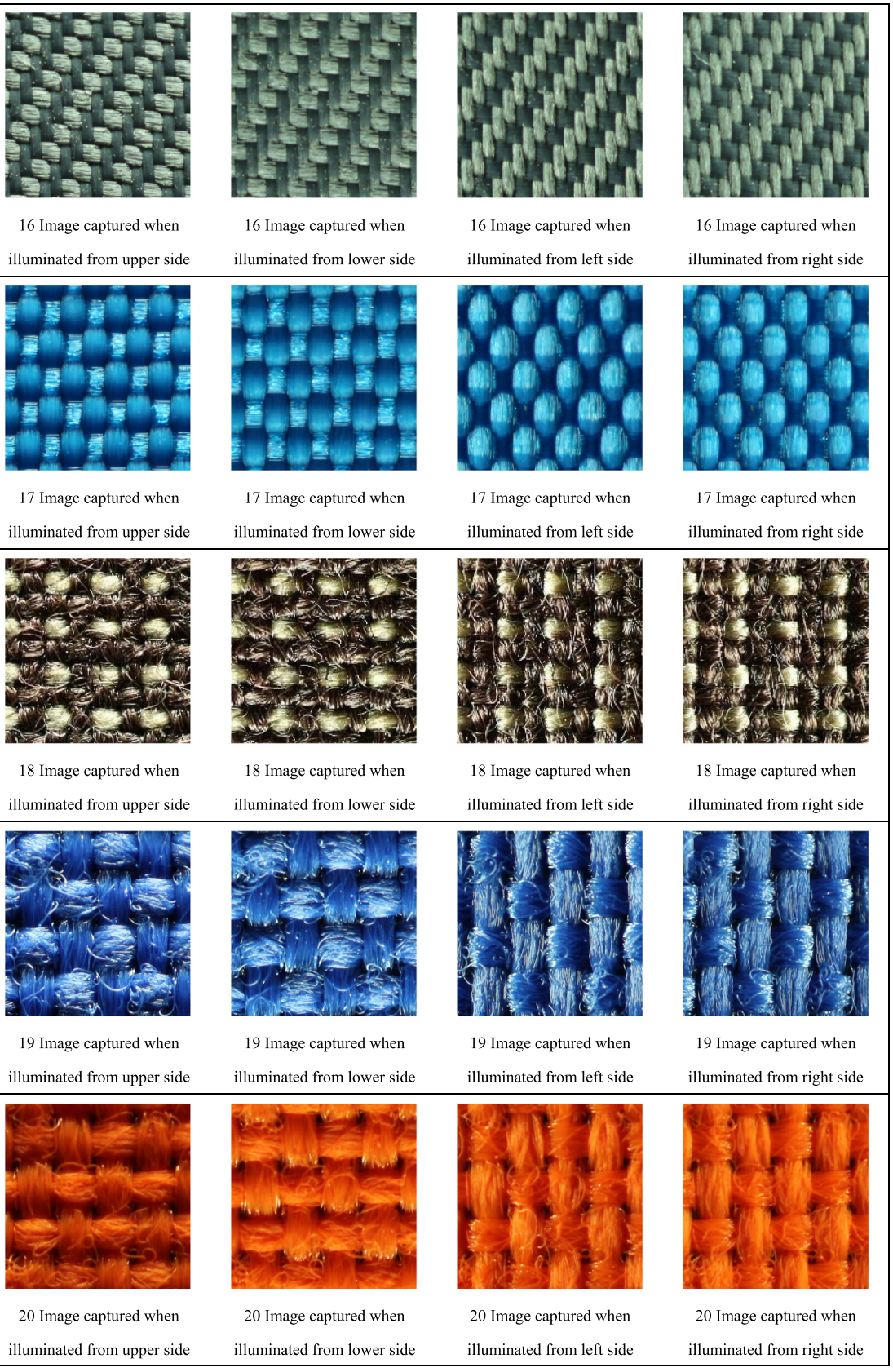


Fig. 3 (*continued*)

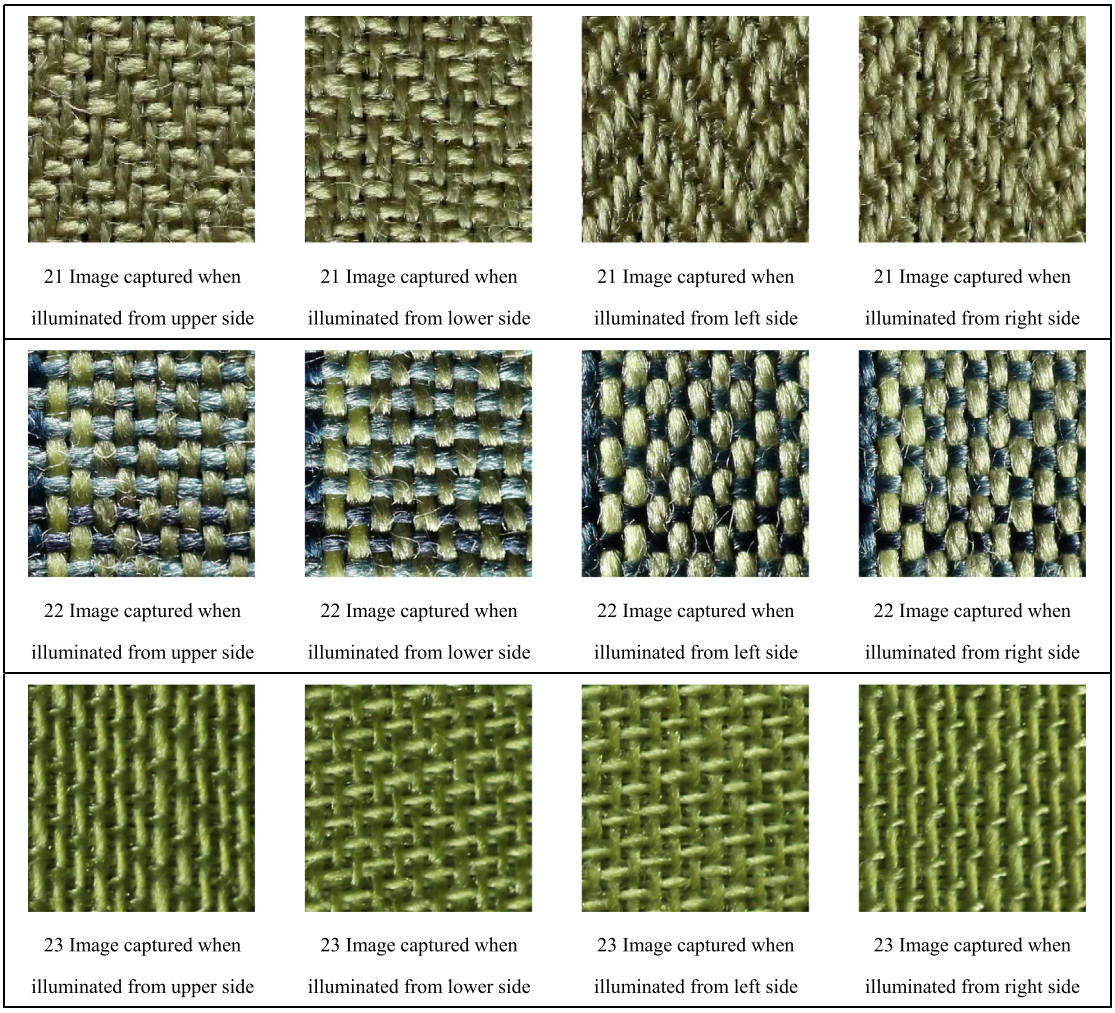


Fig. 3 (*continued*)

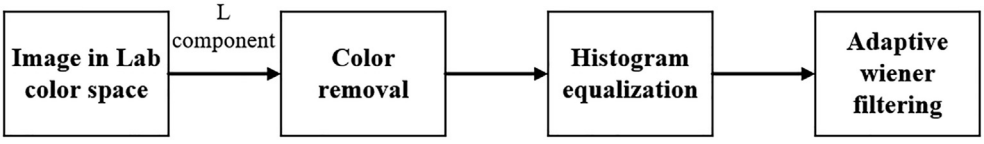


Fig. 4. Flow chart of Image preprocessing.

the interlacing point, were adopted in the experiment. Ultimately, it was very effective that the weave pattern image was generated after obtaining the attributes of all the interlacing points.

In this paper, we proposed a new comprehensively-designed and appropriately-performed method for the weave pattern of woven fabrics, utilizing the concept of photometric stereo imaging and gray scale variance under the X-illumination system. One set of photometric imaging system for the digitalization of woven fabrics illuminated from four different directions was established for the weave pattern reorganization based on photometric differential analysis method, aiming at obtaining more clear and complete information about the fabric structure. This system could obtain clearer image in the single direction of the weft or weft and overcome the effect of uneven illumination on the image results. Under the circumstance of the warp yarn information and weft yarn information not interfering with each other, the weave pattern reorganization will result in their respective directions with higher precision. It is reasonable to improve the accuracy of interlacing point recognition, guiding a new direction for follow-up research on complicated fabric textures.

2. Methodology

To achieve the objectives of photometric differential analysis and weave pattern recognition in a reliable way, twenty-three woven fabrics were selected in this research. By controlling the illumination directions, the images with light sources coming from the upper side, lower side, left side and right side of twenty-three fabric samples were acquired and a total of 92 reflected images were used to subsequent efficient analysis. At the beginning, the images of each fabric sample were registered and cut, and it follows that they were converted into the LAB space, with only the L component of the image extracted and converted into a gray

scale image. The attainment of this objective of the initialization of adaptive mesh model which is used for processing the fused image, requires the fusion of the images captured when illuminated from four directions through the gradient pyramid fusion method after performing the histogram equalization and adaptive wiener filtering. Giving the credit to the adaptive mesh model, the images were divided into a great deal of sub-images of each interlaced point. Based on photometric difference of these images, the gradations

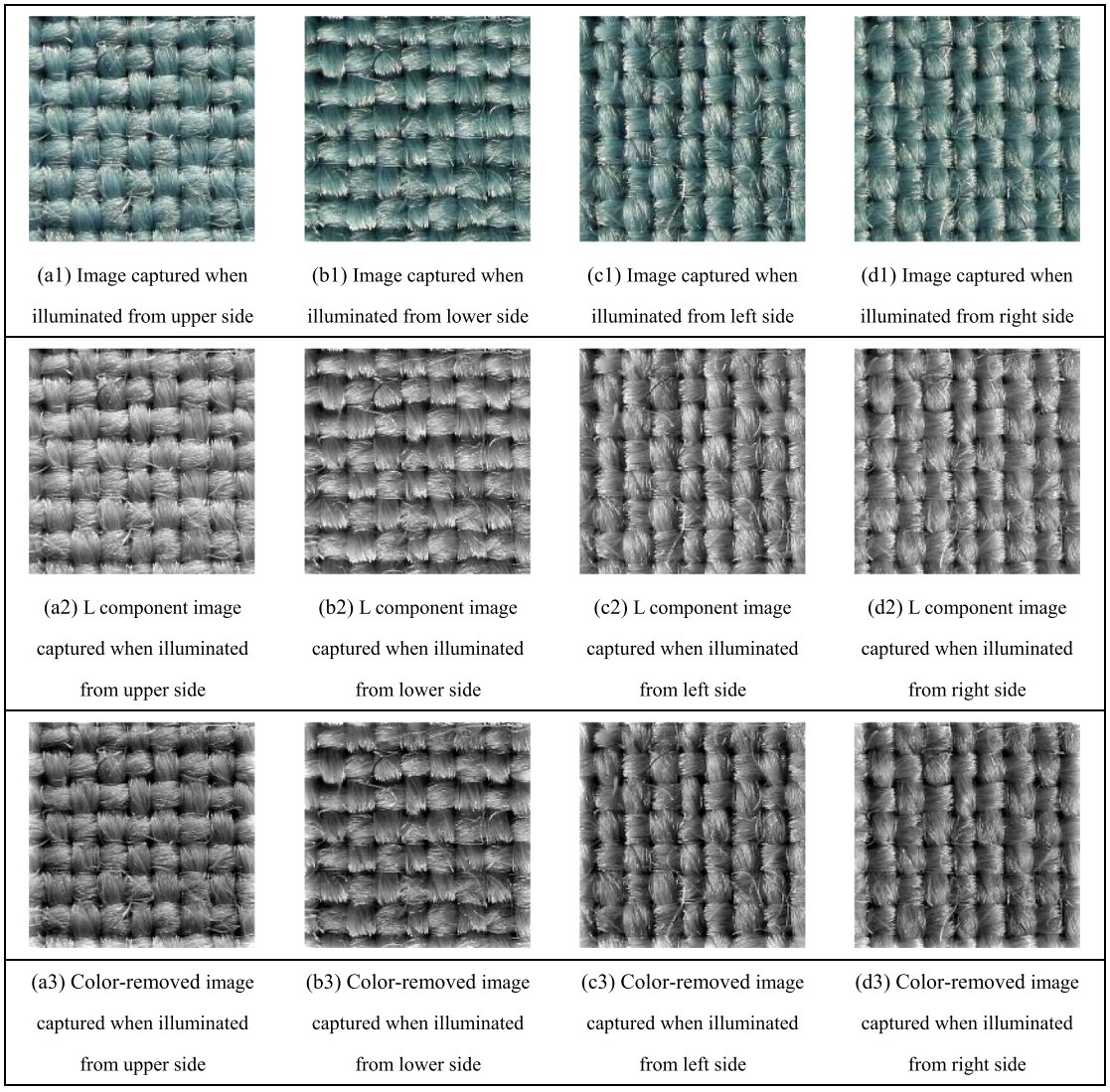


Fig. 5. The images of Sample 1 from four illumination directions.

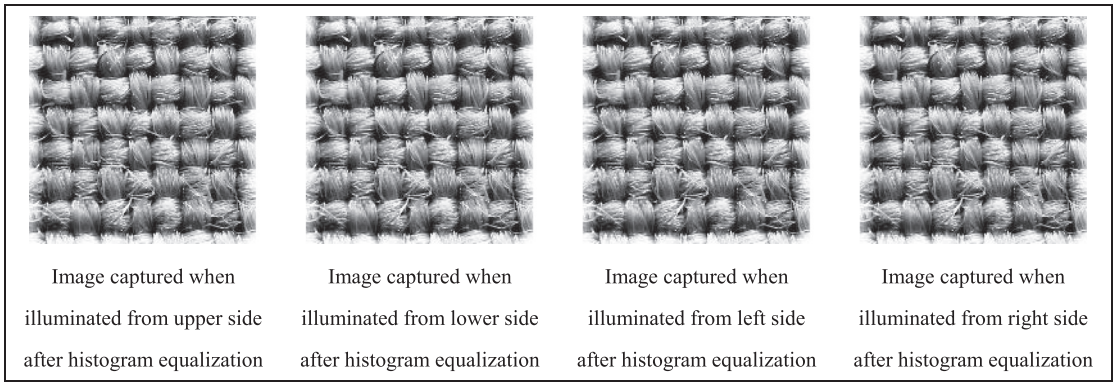


Fig. 6. Images of Sample 1 illuminated from four directions after histogram equalization.

of the convex portions in the images can be used to determine the attributes of all interlacing points, followed by the weave pattern repetition image generated finally (Fig. 1).

2.1. Photometric differential image acquisition system set-up

In this paper, a dedicatedly-designed imaging system equipped with a illumination module (with light sources coming from the upper side, lower side, left side and right side, respectively) was established using a digital camera with intrinsic imaging parameters, which is capable of digitalizing a full set of image sequences by adjusting the illumination directions. The system is comprised of six main components, as shown in Fig. 2:

(1) A high-resolution Canon SLR (EOS 5D Mark II) Camera is used to capture a full set of reflected image sequences under different illumination directions, with the image resolution over 20,000,000 pixels.
(2) A self-designed fabric splint consisting of a strip magnet is employed, for the purpose of fabric mesh alignment.
(3) Four flat-type light sources are placed in a symmetrical manner on the upper, lower, left and right sides of the platform, producing a single illumination source in each direction, respectively. The 6500 K artificial sunlight is selected as the light source, with brightness set to about 1000.
(4) Four self-developed parallel light covers, which can concentrate the light from a single incident source onto the mirrors, are equipped. Coming from four parallel sources, the light is reflected by the mirror onto the fabric surface.
(5) A closed dark box, measuring 45 cm × 45 cm × 45 cm, was used to reduce the influence of other diffuse reflections during the image collection process.
(6) A server with the software (MATLAB R2016b), utilized for processing and analyzing the captured images, is chosen.

2.2. Sample preparation and image acquisition

Developed with an illumination module with light sources coming from the upper side, lower side, left side and right side, respectively, a self-built imaging system was established to obtain the woven fabric images based on the photometric differential analysis in this paper. In our experiments, the image resolution was set to approximately 300 pixels per inch initially; later on, the affine transformation was introduced for image alignment and cropping; as a result of that, the images captured when illuminated from four directions (i.e., upper side, lower side, left side and right side) of twenty-three fabrics were obtained, as shown in Fig. 3, with the size of the finally-aligned images of 512 pixels × 512 pixels.

2.3. Image preprocessing

In the first step during the image preprocessing, the attainment of objectives of noise interference removal and its structural information improvement required the color fabric image to be processed by the adaptive wiener filtering and histogram equalization [22], as shown in Fig. 4. After that, the gray scale projection curves of longitude and latitude were obtained by means of gray scale projection, which was removed by mean filtering [23]. The mean value, used to remove the local maximum value in the curve, was adopted. After determining the curve trough coordinates, the warp and weft positioning, segmentation and organization points were realized [24]. Completed with the help of the interlacing point grid correction method, it was performed based on the local brightness of the weft yarn. Finally, the interlacing point image, containing the complete edge information, was extracted for feature parameter extraction and attribute recognition. The images of the Sample 1 captured when illuminated from four directions are as shown in Fig. 5.

What's more, in order to make the subsequent image fusion faster and more accurate, it is necessary to convert four photometric images under different illuminating direction into LAB space, extract only the L component and remove the color of these images.

2.3.1. Histogram equalization

Established as per the statistical principle, image histogram seems like a graph, representing the gray scale of each level in

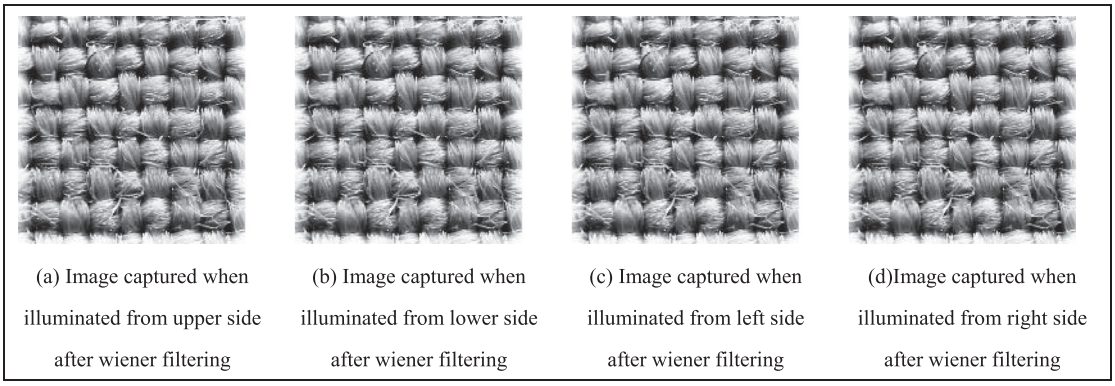


Fig. 7. Images of Sample 1captured when illuminated from four directions after wiener filtering.

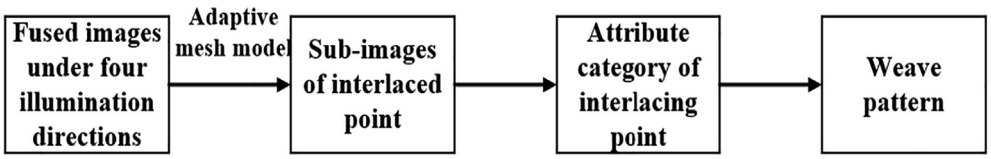


Fig. 8. Flow chart of weave pattern recognition.

the image and its corresponding frequency of occurrence [25]. The abscissa indicates the gray level of different levels of the image, on the other hand, the ordinate denotes the frequency of the corresponding gray scale.

Its calculation formula is as follows:

$$P(k) = \frac{n_k}{N} \tag{1}$$

where N refers to the total number of pixels in the image,n_k stands for the number of pixels with a gray scale of k, and $P(k)$ represents the frequency at which the pixel appears.

Assisted by the gradation transformation, the histogram equalization is a method capable of adjusting the contrast of images in an automatic manner [26], with basic idea intended for determining the gradation transformation function by the probability density of gray scale. By transforming the gray histogram of the original image into a uniformly distributed form, the original image is nonlinearly stretched, and besides, the concentrated gray value is redistributed into the entire gray scale. These two operations are carried out, in order to enhance the contrast in original image. The transformation formula of the transformation function $T(r)$ and the original image probability density function $p_r(r)$ is as below:

$$T(r) = \int_0^r p_r(r)dr \quad (0 \le r \le 1) \tag{2}$$

Histogram equalization processing is performed, on the basis of the color fabric image. No matter it is a plain weave or a twill weave, the gray value distribution of the image after the color removal appears relatively concentrated and the overall image exhibits dark, with comparatively-low contrast. After the histogram equalization processing, the variation range of the image gradation becomes larger and the frequency fluctuation of each gradation gets smaller, with more visible image contrast, as a result of that, the structure of the tissue point presents clear. Histogram equalization images of Sample 1 are as shown in Fig. 6.

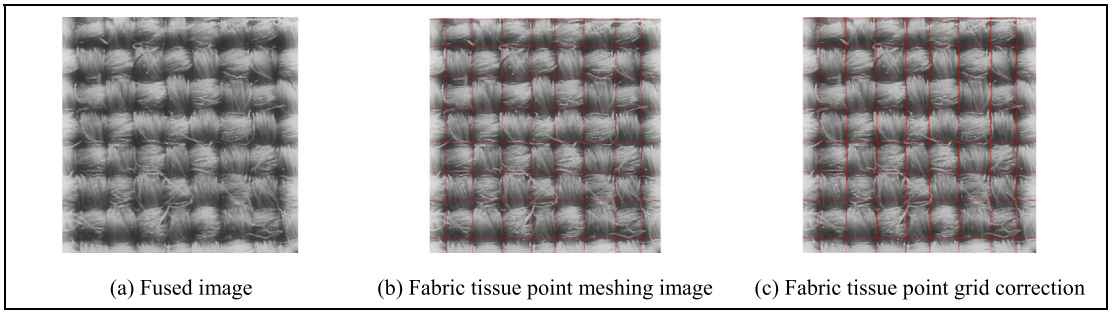


Fig. 9. Fused image for meshing.

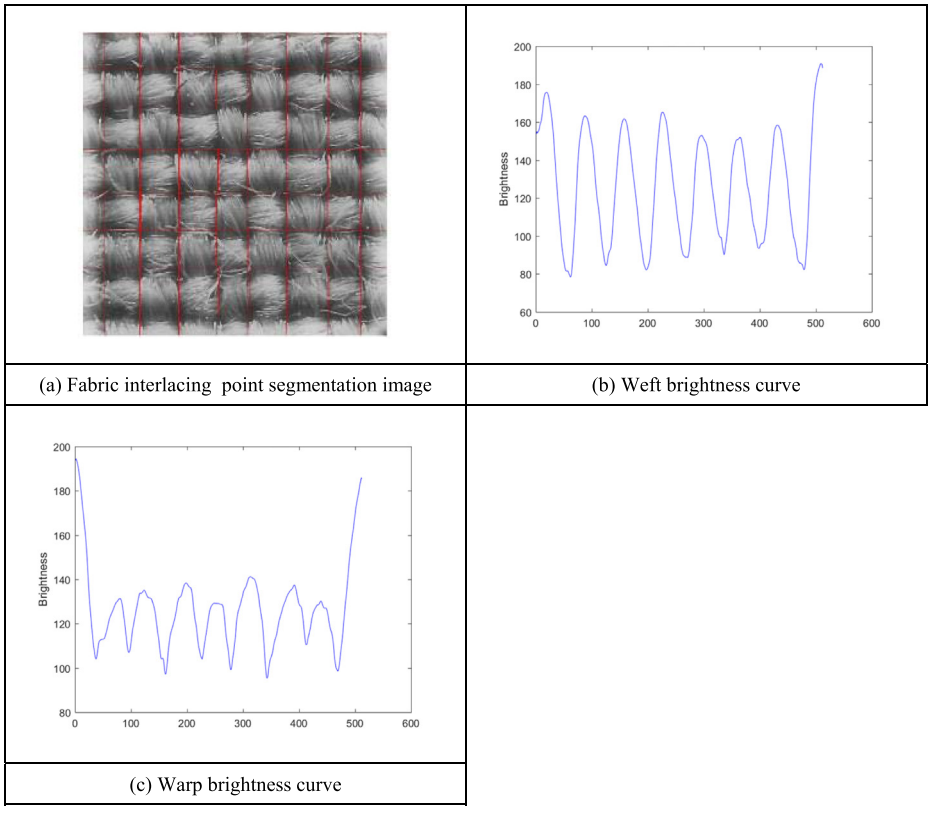
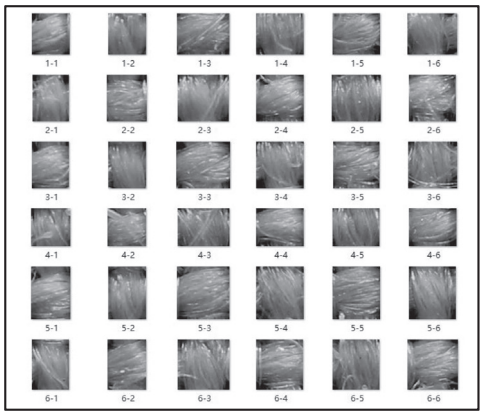


Fig. 10. Corrected interlacing point grid segmentation figure and brightness curve.

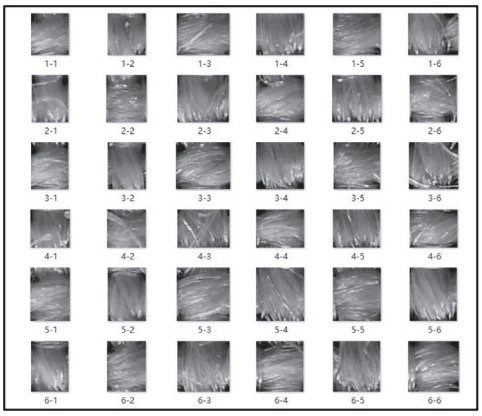
2.3.2. Adaptive filtering denoising

In the process of acquisition and transmission of digital images of fabrics, a large amount of random noises occur, inevitably, influenced by hardware equipment and environmental factors. The presence of noise will cause the interference with the image information, to a certain degree, hindering follow-up image processing and analysis. Noise removal from the fabric image, therefore, is required to be carried out in advance. In this paper, the image was denoised by the application of adaptive wiener filtering.

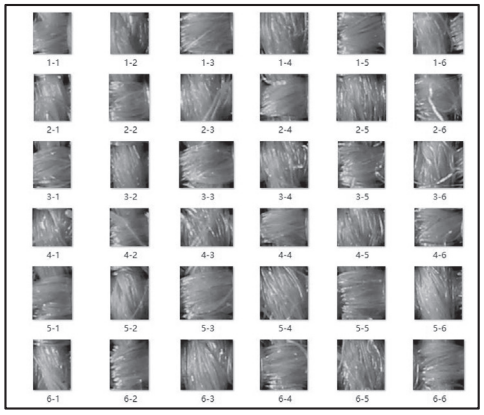
Acting as a kind of two-dimensional & non-linear adaptive filtering method, the wiener filtering is conducted based on estimating the neighborhood pixel values [27]. By calculating the variance in the local region of the image, the region will be smoothed slightly under the condition of large regional difference; on the contrary, more smoothing processing will be required in the case of small regional difference. Therefore, compared with other linear methods, the wiener filtering method exhibits the advantage of preserving the image edge and other high-frequency components



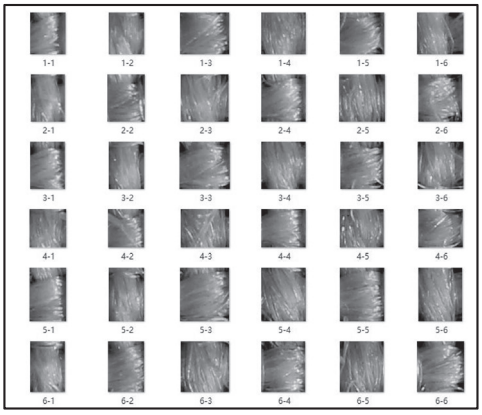
(a) All sub-images of interlacing point (illuminated from upper side)



(b) All sub-images of interlacing point (illuminated from lower side)



(c) All sub-images of interlacing point (illuminated from left side)



(d) All sub-images of interlacing point (illuminated from right side)

Fig. 11. All sub-images of the interlacing point (illuminated from four directions).

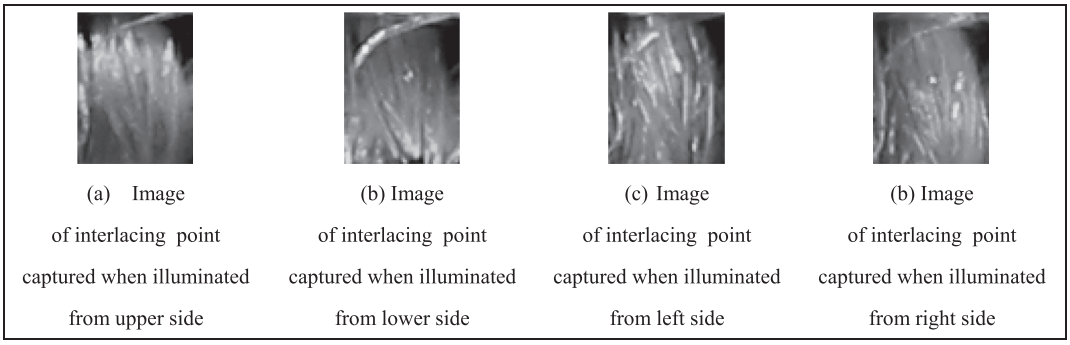


Fig. 12. Images of interlacing point captured when illuminated from four directions (No. 4–6).

while filtering out the noises, providing preferable noise reduction effect and superior image adaptation effect.

The principle of the wiener filtering is as follows:

First, the local mean μ and variance σ^2 of each pixel should be estimated:

$$\mu = \frac{1}{MN} \sum_{n_1, n_2 \epsilon \eta} \alpha(n_1, n_2) \tag{3}$$

$$\sigma^2 = \frac{1}{MN} \sum_{n_1, n_2 \epsilon \eta} \alpha^2(n_1, n_2) - \mu^2 \tag{4}$$

where η indicates $M * N$ as the domain, afterwards, the wiener filtering is built using the local mean and variance. The filter transformation function is as follows:

$$b(n_1, n_2) = \mu + \frac{\sigma^2 - v^2}{\sigma^2}(\alpha(n_1, n_2) - \mu) \tag{5}$$

where v^2 denotes the noise variance, in general, it is of a low value. Therefore, the correspondingly low variance value of the region can be acquired in the situation of certain noises existing in the smooth region of the image, and at that time, the wiener filtering performs more smoothing processing in that region and the noises will be removed. If the area in the fabric image where the structural information such as the boundary of the yarn is located possesses a large variance value, only a little smoothing will be executed by the wiener filtering in that area to preserve the structural features. The images of Sample 1 processed by the wiener filtering are as shown in Fig. 7.

2.3.3. Fusion of images captured when illuminated from four directions

During the fabric structure identification, the yarn edge information should be highlighted. Due to the close arrangement of light and yarn during the image acquisition, nevertheless, the blurred and indistinct yarn edge will appear. In order to solve this problem, the image sharpening which can be generally classified into two categories (differential method and high-pass filtering) should be implemented to improve the edge and contour information. Within a small pixel distance, the gray value between the yarn gaps is significantly changed, for this feature, the high-pass filtering can be used to sharpen the yarn edge.

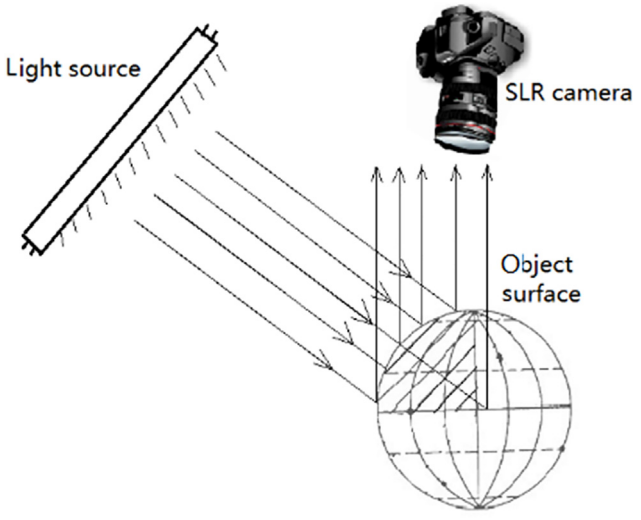
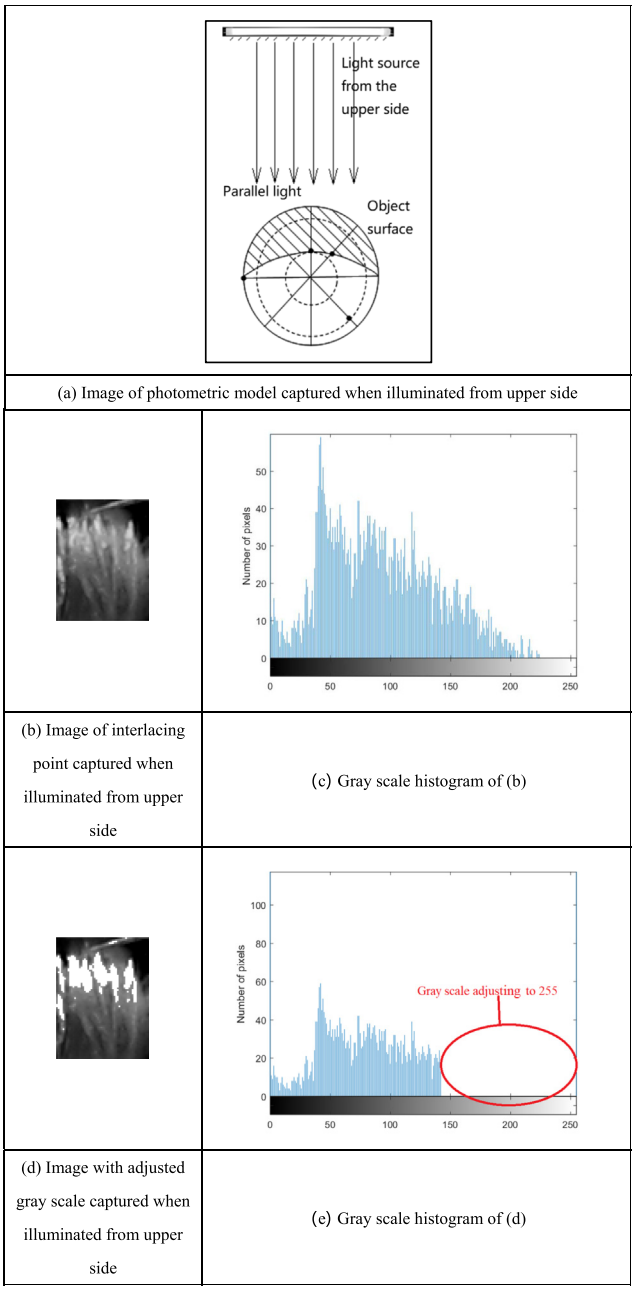


Fig. 13. Photometric model of interlacing point.

For the sake of laying the foundation for the further division of the organization point grid lines, there must be the combination of the four metering images together with the extraction of common yarn edge information.

Image fusion algorithm, based on the Gradient Pyramid (GP) decomposition, is a type of multi-scale decomposition algorithm [28] with its image representation obtained by performing a gradient operator on each layer of the Gaussian pyramid image. Each layer of GP decomposition image contains detailed information in four directions of horizontal, vertical and two diagonal lines, able to extract the image edge information and improve the stability and noise resistance with much better results. And besides, the directional gradient pyramidal decomposition provides good directional edge and detailed information for the image.



(a) Image of photometric model captured when illuminated from upper side

(b) Image of interlacing point captured when illuminated from upper side

(c) Gray scale histogram of (b)

(d) Image with adjusted gray scale captured when illuminated from upper side

(e) Gray scale histogram of (d)

Fig. 14. Adjusted image of interlacing point (No. 4–6) and image with adjusted gray scale captured when illuminated from upper side.

1) Image gradient pyramid decomposition

By performing gradient direction filtering on each decomposition layer (except the highest layer) of the image Gaussian pyramid, the gradient tower decomposition can be obtained as below:

$$D_{LK} = d_K \cdot (G_L + w_0 \cdot G_L), 0 < L < N, K = 1, 2, 3, 4 \quad (6)$$

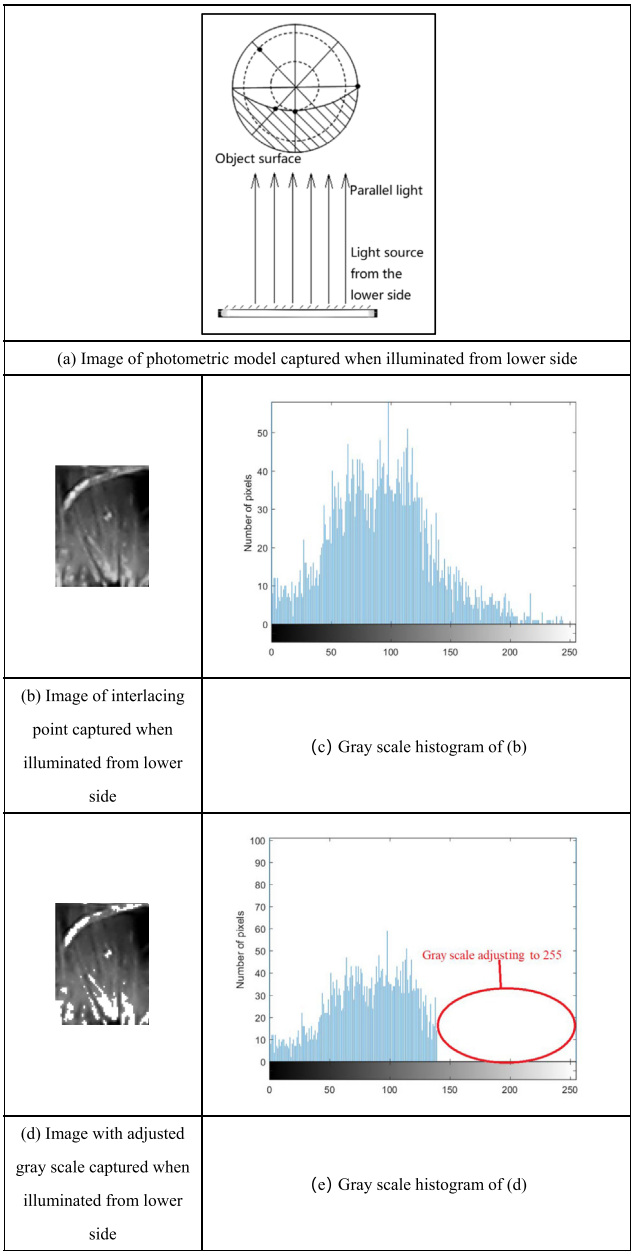
Herein, $\cdot$ denotes a convolution operation; D_{LK} represents the L layer in k direction gradient tower image; G_L refers to the L layer image of the image Gaussian pyramid; d_K stands for the k direction gradient filter operator, defined as follows:

$$d_1 = [1 - 1]; \quad d_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}; \quad d_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \quad d_4 = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad (7)$$

The directional gradient filtering is performed on each layer of the image Gaussian pyramid through d_1, d_2, d_3, d_4, and it follows that four decomposition images including horizontal, vertical and two diagonal directions are obtained on each decomposition layer (except the highest layer). Not only the gradient pyramid decomposition of the visible image is performed on a multi-scale & multi-resolution basis, but also each decomposition layer is composed of images containing detailed information from four directions.

2) Reconstructed image

After respectively merging each layer of the pyramid image, it is necessary to reconstruct the original image based on it. Introduced as an intermediate result, the FSD Laplacian pyramid must be adopted, that is, the gradient pyramid is converted into a Laplacian pyramid followed by the reconstructed original image by the Laplacian pyramid, with its construction process shown as follows [29].



(a) Image of photometric model captured when illuminated from lower side

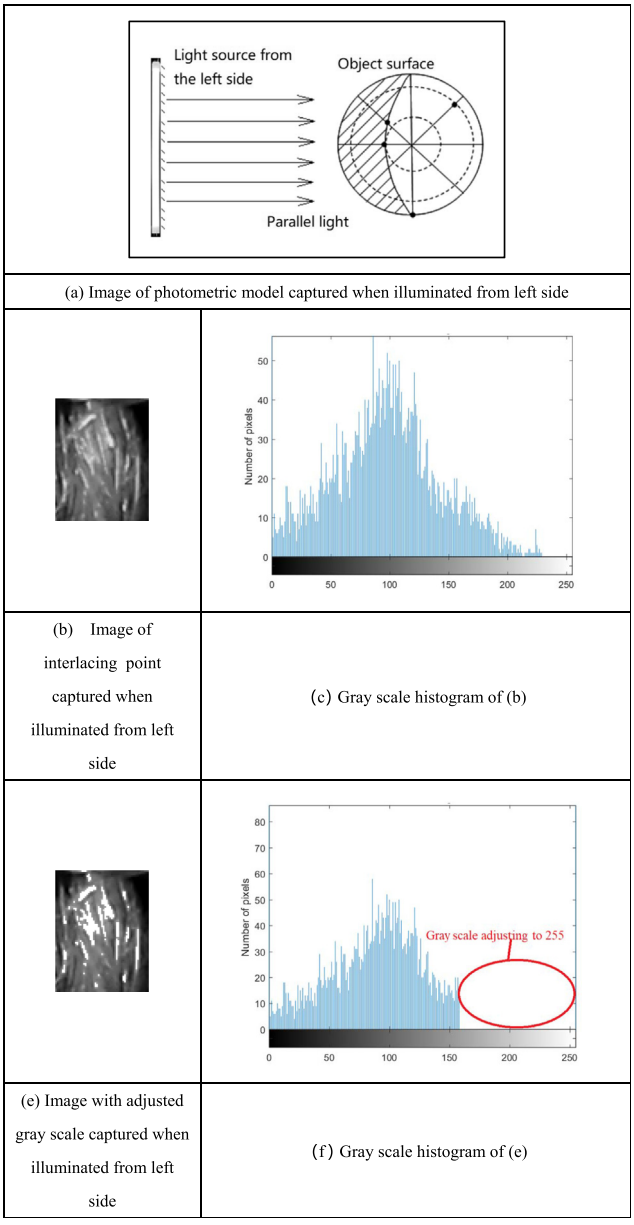
(b) Image of interlacing point captured when illuminated from lower side

(c) Gray scale histogram of (b)

(d) Image with adjusted gray scale captured when illuminated from lower side

(e) Gray scale histogram of (d)

Fig. 15. Image of interlacing point model (No. 4–6) and image with adjusted gray scale captured when illuminated from lower side.



(a) Image of photometric model captured when illuminated from left side

(b) Image of interlacing point captured when illuminated from left side

(c) Gray scale histogram of (b)

(e) Image with adjusted gray scale captured when illuminated from left side

(f) Gray scale histogram of (e)

Fig. 16. Image of interlacing point model (No. 4–6) and image with adjusted gray scale captured when illuminated from left side.

The directional gradient pyramid should be converted into the FSD Laplacian pyramid, with the image of the L layer of the FSD pyramid made to be L_L, then

$$L_L = \sum_{k=1}^{k=4} -\frac{1}{8} d_k \cdot D_{IK} \tag{8}$$

The FSD Laplacian pyramid image is transformed into a Laplacian pyramid image:

$$LP_L = (I + w) \cdot L_L \tag{9}$$

The G_L interpolated value is amplified so that the enlarged image size is same as the G_{L-1} size, and the Expand operator is introduced. Expand is same as the G_x^* definition.

$$G_N = LP_N, when L = N \tag{10}$$

$$G_L = LP_L + Expand(G_L + 1), when 0 \le L < N \tag{11}$$

The original image G_0 can be reconstructed by sequentially making $L = N-1, N-2, \cdots, 0$ in Formula (15) which is

$$G_L = LP_0 + Expand(LP_1) + Expand(Expand(LP_2)) + \cdots + Expand(Expand(LP_N)) \tag{12}$$

3. Experiment results and discussion

3.1. Fabric interlacing point meshing

As a reflected light image, in nature, the fabric image was acquired. Since the warp and weft yarns are in a buckling and interlacing state, the axial center of the yarn floating on the fabric surface would present the maximum brightness, in the event of the fabric surface irradiated with reflected light. Both sides of the axial line at which away from itself shows a gradual decreasing trend, and the brightness reaches a minimum at the yarn gap. According to the brightness characteristics of the yarn, a larger gray value at the axis line together with a smaller gray value at the gap of the yarn pixel points in the reflected light image can be gained. Therefore, the division and positioning of the yarn could be realized according to the gradation variation characteristics of the yarn image (Fig. 8).

Fused image for meshing is as shown in Fig. 9.

According to the brightness characteristics of the yarn, the peak position in the latitudinal gray scale projection curve of the fabric corresponds to the warp axis, at the same time, the trough position corresponds to the warp gap. In view of this, by determining the trough coordinates, the warp gap position can be positioned to realize the warp yarn splitting. In the same way, by determining the trough coordinates in the longitudinal gray scale projection curve of the fabric, the position of the weft gap can be located to achieve weft splitting.

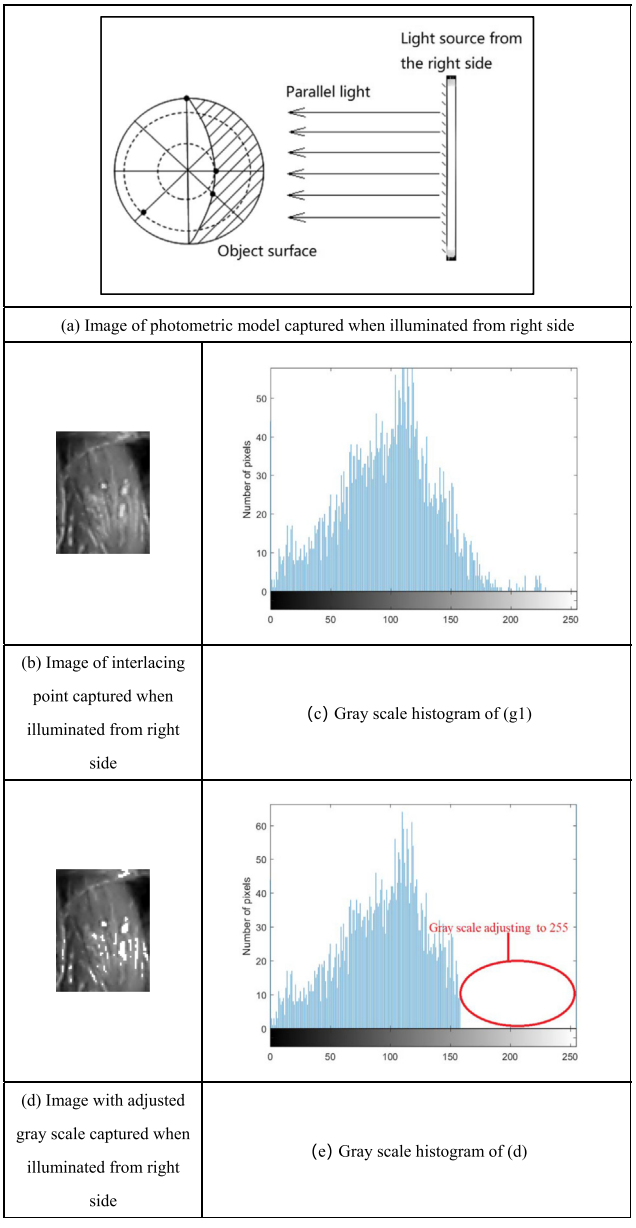
3.2. Warp and weft yarn positioning and segmentation based on grey scale projection method

By using the latitude and longitude gray scale projection curves of the fabric image, its trough coordinates can be determined. Their positioning plus coordinates of the warp and the weft yarns could be realized, respectively. What's more, the interlacing point grid initialization could be completed. The corrected interlacing point grid segmentation figure and brightness curve image are as shown in Fig. 10.

Based on the local brightness of the weft yarn, the interlacing point grid correction was completed. Finally, the interlacing point image containing the complete edge information was adopted for feature parameter extraction and attribute recognition. All sub-images of the interlacing point are as shown in Fig. 11.

3.3. Interlacing point attribute recognition

In the woven fabric, the warp and weft interlacing points exhibit in a state of mutual floating due to the interlacing of the warp and weft yarns. In the event of the warp yarn floating above the weft yarn, the warp point will be regarded as interlacing point. On the contrary, the weft point will be deemed as interlacing point when the weft yarn floats above the warp yarn. In the reflected light image of the fabric, from different interlacing points comes diverse gray information. The gray scale change rule at the interlacing point can be mainly classified into two categories: gray gradient and gray scale variation. For the warp interlacing points, the gray value of the pixel along the warp direction (vertical direction) firstly increases, and then decreases in the interlacing point image, finally reaches the maximum at the warp and weft interlacing



(a) Image of photometric model captured when illuminated from right side

(b) Image of interlacing point captured when illuminated from right side

(c) Gray scale histogram of (g1)

(d) Image with adjusted gray scale captured when illuminated from right side

(e) Gray scale histogram of (d)

Fig. 17. Image of interlacing point model (No. 4–6) and image with adjusted gray scale captured when illuminated from right side.

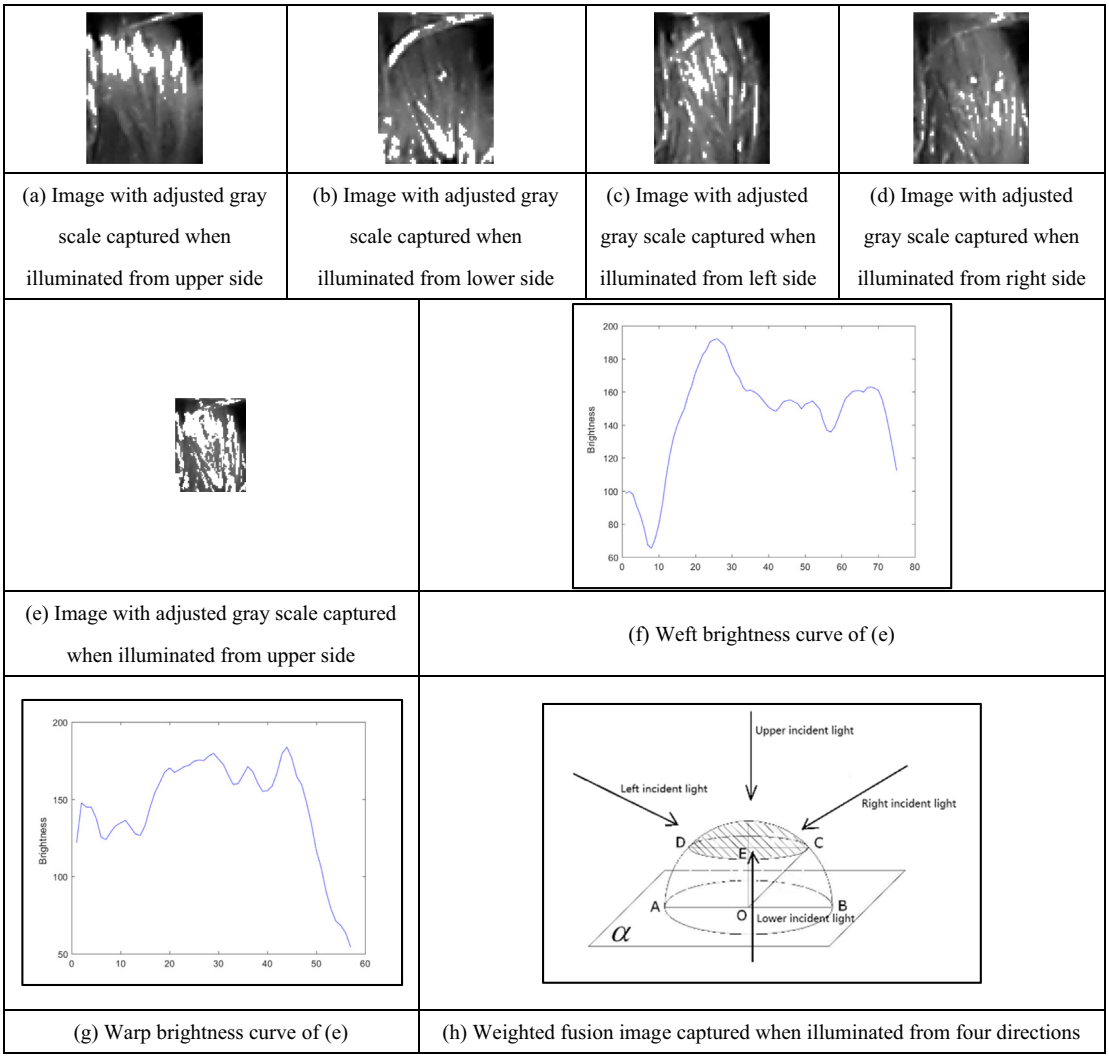


Fig. 18. Image of interlacing point with adjusted gray scale captured when illuminated from four directions (No. 4–6) and its brightness curve images.

point; along the weft direction (horizontal direction), due to the weft yarn sinks below the warp yarn, the a sudden change of gray value of the pixel will happen at the interlacing point, reaching the smallest. Compared with the weft interlacing points, an opposite change of the gray scale in the image of the warp interlacing point would be found. The gradation makes a change all of a sudden in a dramatic way along the warp direction, however, the gradation changes gradually along the weft direction.

In accordance with the two completely opposite gradation transformation rules in both the warp and weft interlacing point images of woven fabrics, this paper proposed a novel digital method based on photometric differential analysis, for the sake of identifying the attributes of interlacing points. The images of the interlacing point captured when illuminated from four directions (No. 4–6) are as shown in Fig. 12.

In order to describe, in an objective form, the gray scale gradation and gray scale abrupt features in the warp and weft interlacing point images, both horizontal and vertical gray scale mean changes were selected as feature parameters to represent the interlacing point image. The mean value of horizontal changes which reflects the gray level variation at the interlacing point in the horizontal direction can be obtained. With larger value, comes greater grayscale variation in the horizontal direction and more obvious gray scale variation. The mean value of vertical changes, reflecting the gray scale variation at the interlacing point in the vertical direction, can be achieved by calculation. The larger the value is, the larger the gray scale variation in the vertical direction is and the more obvious the gray scale variation is (Fig. 13).

Supposing that M represents the number of pixels in the vertical direction of the interlacing point image, N denotes the number of pixels in the horizontal direction of the image, $G(x,y)$ indicates the gray value of the image at (x,y), d refers to the pixel between the two pixels. The width of the horizontal gray scale variation of the image is expressed as follows:

$$Hcov(d) = \frac{1}{M \times (N-d)} \sum_{y=1}^{M} \sum_{x=1}^{N-d} |G(x,y) - G(x+d,y)| \tag{13}$$

The vertical gray scale variation of the image is expressed as follows:

$$Vcov(d) = \frac{1}{N \times (M-d)} \sum_{y=1}^{M-d} \sum_{x=1}^{N} |G(x,y) - G(x,y+d)| \tag{14}$$

In short, the gray scale of an image could reflect the surface contour of an object. If illuminated, much higher luminosity of the raised area than any other parts would be obtained. By the method of gray scale statistics, the highlight region exhibits white light through adjusting the image captured when illuminated from the upper side, and finally the image of the interlaced point with

Table 1
Gray scale mean value of interlacing point (No. 4–6) in horizontal direction.

Gray scale mean (Horizontal)	Columns 1	Columns 2	Columns 3	Columns 4	Columns 5	Columns 6
Row 1	17.2045	11.4575	14.8000	11.8256	16.8606	10.2351
Row 2	9.83730	14.9900	9.4836	16.1182	12.6988	17.2473
Row 3	16.9817	10.4645	17.9654	10.9671	18.2901	10.3289
Row 4	13.4901	17.3022	12.0648	19.5210	12.9878	17.7376
Row 5	16.7601	10.1510	15.6807	10.8678	16.3772	11.5192
Row 6	12.1033	16.9677	12.4417	19.6468	15.2297	17.0050

Table 2
Gray scale mean value of interlacing point (No. 4–6) in vertical direction.

Gray scale mean (Vertical)	Columns 1	Columns 2	Columns 3	Columns 4	Columns 5	Columns 6
Row 1	10.0162	12.8796	10.6824	15.8755	10.8860	14.9477
Row 2	12.0812	12.7566	14.4699	9.5327	15.8524	13.8357
Row 3	12.7534	13.0751	10.2495	15.3895	11.4548	13.5477
Row 4	15.4464	11.9808	16.1238	12.5506	15.1550	10.3661
Row 5	11.8562	14.9798	12.7489	16.6074	10.5924	16.4426
Row 6	17.4204	12.5521	15.2775	11.3521	17.1656	13.3790

obvious luminosity difference can be obtained, as shown in Fig. 14. The images of the interlacing point captured when illuminated from the upper side, lower side, left side and right side with obvious luminosity difference, in the same way, could be obtained, as shown in Figs. 15–17.

In the photometric stereoscopic imaging system, the relative position between the camera and target remains unchanged. Under different illumination conditions, only gray scale variations happen to the images, without any geometric deviation [30]. Therefore, it is possible to solve the gradation changes occurring in different images by the same pixel. According to the design principle of Lambertian surface reflection model, the image gray scale depends on the angle between the target surface normal and the light source. For conventional photometric stereo vision imaging systems, the light source is capable of providing only the intensity information rather than positional information for the image pixels, which are provided by the area unit of the camera. By extension, it could be understood that the gray scale of the image relies on the angle between the target surface normal and the non-position encoding element. In this properly-improved photometric stereo vision system, the relative position between the camera and object remains unchanged, so that the gray scale difference of the images captured when illuminated from four directions be only determined by the illuminating direction of a single incident light.

The highlight regions in the images captured when illuminated from four directions could be concentrated, furthermore, the non-highlight regions of these images were gradated by weighted averaging. In view of this, an image with a completely-adjusted raised highlight region could be obtained. By analyzing the highlight regions of protrusions in the image, the polarities of the gray scale differences in the vertical and horizontal directions were analyzed to determine the warp and weft properties of the interlacing points in the image, as shown in Fig. 18 and Tables 1 and 2.

After determining the parameters of the interlacing point, both x-axis and y-axis were taken as the mean value of horizontal change and vertical change, respectively. The classification coordinate system shown in Fig. 20 was established, and also, the interlacing point attribute identification based on the photometric difference was obtained. The specific process of the method is as follows (Fig. 19):

After performing the gray scale variation, the mean value of horizontal gray scale variation (H) and the mean value of vertical gray scale variation (V) of the interlacing point were calculated, followed by the comparison between these two numerical values.

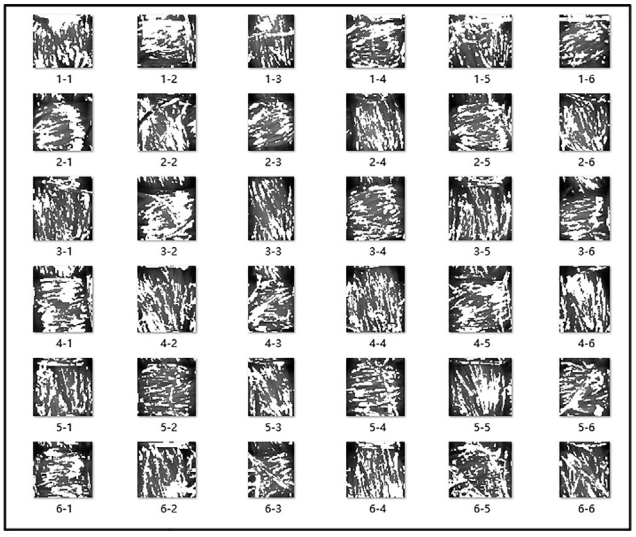


Fig. 19. Image of interlacing point with adjusted gray scale captured when illuminated from four directions (No.4–6) and its brightness curve images.

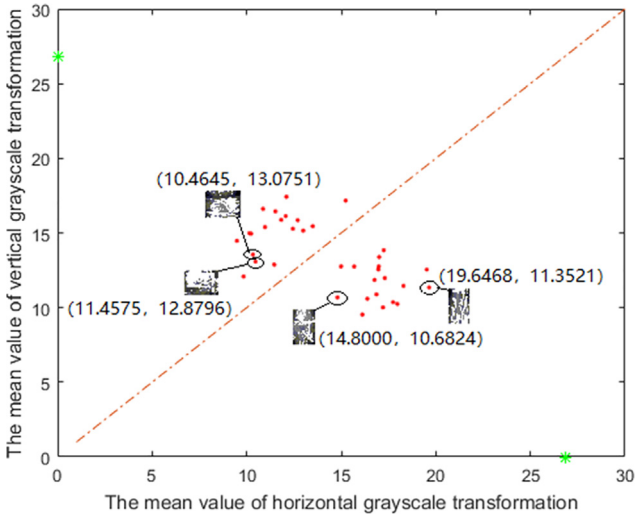
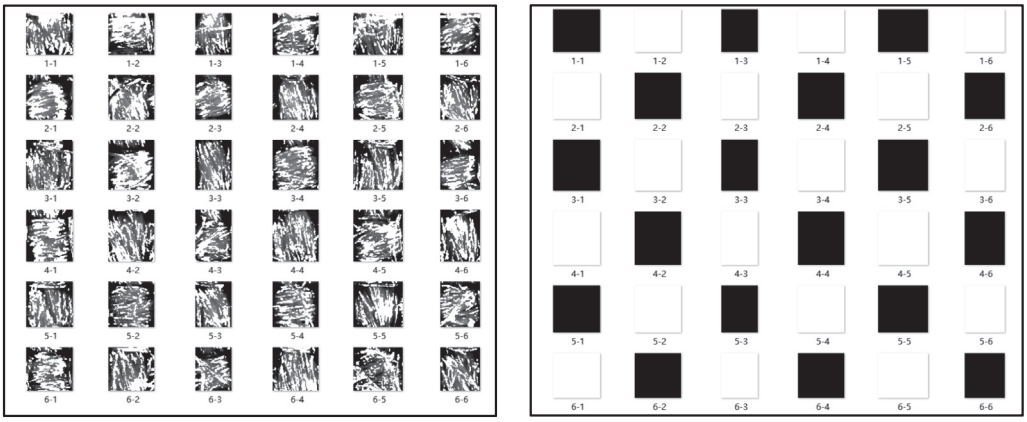


Fig. 20. Image of interlacing point with adjusted gray scale captured when illuminated from four directions (No. 4–6) and its brightness curve images.

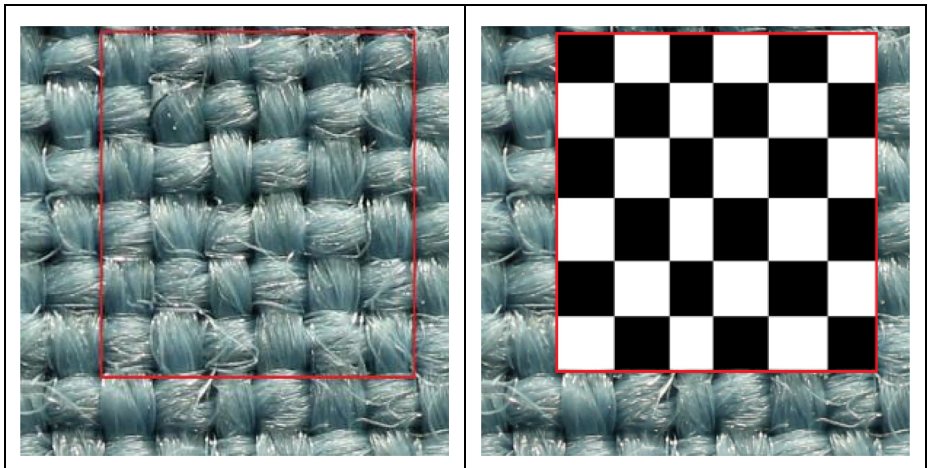


(a) Interlacing point images

(b) Recognition result of (a)

(c) Weave pattern

Fig. 21. Identification results of interlacing points.



(a) Original image of sample 1

(b) Pattern repetition of sample 1

Fig. 22. Pattern repetition of sample 1.

Under the condition of the value of H higher than that of V which belongs to the horizontal gray scale variation, it demonstrates the larger gray scale variation of the interlacing point in the horizontal direction, with interlacing point classified as the warp interlacing point. Belonging to the vertical gray scale variation, if the value of V is greater than that of H, it means larger gray scale variation of the interlacing point in the vertical direction exists, with interlacing point classified as the weft interlacing point.

The recognition was performed, in the same way, on all the interlacing points, and thus, the interlacing point classification result was obtained. Therefore, the weave pattern image of the fabric could be finally generated, as shown in Fig. 21.

The recognition process was performed, likewise, on the samples stored in the fabric sample library based on the same photometric difference analysis method. Firstly, the pixel-level alignment plus cutting was performed on all images. And then

the yarn edge was strengthened by the fusion algorithm and sharpening algorithm, followed by the grid line drawn based on the yarn edge in the fused image. Helped by the same adaptive mesh model, the images can be divided into a large quantity of sub-images of each interlaced point based on gray-scale-based projection algorithm. Based on the principle of appropriately-improved

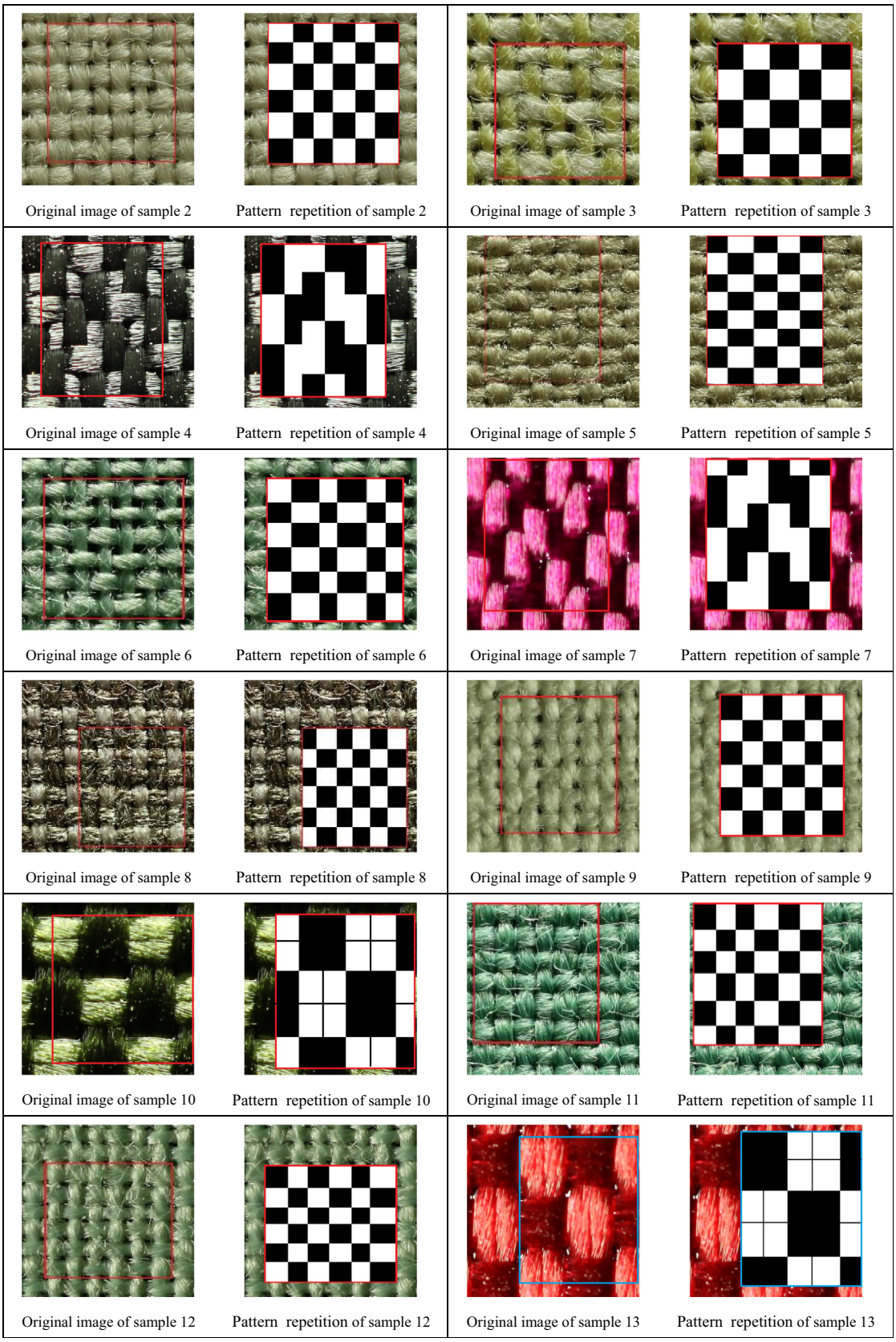


Fig. 23. Pattern repetition of the other woven fabrics.



Fig. 23 (*continued*)

photometric stereo vision, the gradation adjustment of the sub-images of interlacing point in the images was respectively performed, that was, the gradation increased to 255 for the protrusion, with this region highlighted. Finally, the gradation of the highlight region in the image was extracted and aggregated, and the gradation of the non-highlight region was weighted and averaged. It is of great necessity to obtain an image with a completely-raised highlight region, that is mainly because the polarity of the gray scale variation in the vertical and horizontal directions can be analyzed from the image with the completely-raised highlight area. The values of H and V of the interlacing point were calculated, it followed by the comparison between these two values. The values, thus, determined the warp and weft attributes of the interlacing points in the image. Finally, weave pattern images of other fabrics reserved in the fabric sample library were generated (Figs. 22 and 23).

This method is suitable to most types of fabric, like reverse twill fabric (sample 21), twill fabric (sample 16), Satin fabric (sample 23), buckskin fabric (sample 4 and 7), basket fabric (sample 10 and 13) and plain fabric.

4. Conclusion

Based on the photometric differential analysis, a new dedicatedly-designed and well-developed method was proposed

for the weave pattern analysis in this paper. In particular, the reflected images of the fabric can be acquired digitally from different illumination directions by a self-built imaging system equipped with an illumination module with light sources coming from four directions. By controlling the illumination direction (upper side, lower side, left side or right side), the images of the fabric appearance can be obtained for following efficient analysis. After the histogram equalization and adaptive wiener filtering, the attainment of apparent edge contour information of warp and weft yarns required the image fusion under the condition of being illuminated from one direction through the gradient pyramid fusion method. To achieve the objective of determining yarn alignment in the warp and weft directions of fused image, there must be an adaptive mesh model. According to the design principle of adaptive mesh model, the images can be divided into large numbers of sub-images of each interlaced point based on gray-scale-based projection algorithm. The gradation of the raised portion in the sub-images was adjusted based on the specially-improved photometric difference of images, with adjusted region marked as a highlight region. The finally-processed images were obtained after the other non-highlight region was processed through the weighted average. Both the highlight region and non-highlight region, determining the attribute category of the interlacing point, were used in this experiment. It is very effective that the weave pattern diagram was generated after obtaining the attributes of all the interlacing points.

The experimental results showed that almost 97.22% of recognition accuracy of weave pattern can be gained. Twenty-three woven fabric samples can be successfully identified for their weave patterns, proving that the proposed method is of novel, evolutionary and robust. A photometric differential method being used in fabric structure analysis, exhibits a variety of practical significances. In fact, the warp and weft yarn information will interfere with each other and affect the final identification result. In this case, by controlling the illumination direction, not only the images captured when illuminated from the upper side and lower side in the warp direction but also the images captured when illuminated from the left side and right side in the weft direction with clear image effect can be obtained. It is reasonable to improve the accuracy of interlacing point recognition, guiding a new direction for follow-up research on complicated fabric textures.

However, the limitation of this method is hard to identify the weave pattern of some special fabrics, which refers specially to fabric with apparent interstices between the warp or weft yarns. The operating efficiency of this method also need to be improved.

CRediT authorship contribution statement

Jiaping Li: Writing - original draft , Investigation, Methodology, Writing - review & editing. **Wendi Wang:** Data curation. **Na Deng:** Supervision, Investigation. **Binjie Xin:** Conceptualization, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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